

Chapter 1

1. With a neat diagram, explain the sequence of system programs involved from writing source code to executing a program.
2. Explain different types of text editor in detail.
3. Compare Compiler and Interpreter.
4. Compare Application Software and System Software.

Chapter 2

1. Explain with the help of flowchart the working of Pass I assembler along with databases used.
2. Discuss with example 'forward reference'.
3. State and explain the types of assembly language statements with examples.
4. List and describe the advanced assembler directives with suitable example.
5. Explain with the help of a flowchart, the first pass of two-pass macro processor.
6. Numerical on Pass I and Pass II
7. Explain with flowchart design of two pass assembler.
8. Explain the elements of Assembly Language programming.

Chapter 3

1. Explain Macro and Macro Expansion with example.
2. Explain different features of macros with suitable example.
3. Explain working Single Pass macro processor with neat flowchart.
4. Explain Macro calls within Macro with suitable example.

Chapter 4

1. Explain different functions of loader in detail.
2. Explain Direct Linking Loader with Suitable example.
3. Discuss the concept of "Relocation" in loaders. Explain the various methods of handling relocation during the loading process with example.
4. Explain Relocating Loader in detail.
5. What is the fundamental process of a loader? Explain dynamic loading in detail.

6. Define Relocation and Linking. Why are they crucial functions of a linking loader.

Chapter 5

1. Find out FIRST and FOLLOW for the given grammar.
2. Design a Predictive Parser or LL(1) for the given grammar.
3. What are the phases of compiler with example.
4. Consider the grammar and eliminate left recursion.
5. Explain the operator precedence parser with a suitable example.
6. Short note on :1] Syntax Directed Definition 2] YACC
7. What are tokens, lexemes, and patterns in lexical analysis? Give examples.

Chapter 6

1. Differentiate between Machine-Dependent and Machine-Independent code optimization techniques. Give two distinct examples for each category.
2. Explain design issues in code generation in detail.
3. Explain the concept of basic blocks and flow graph with example the three-address code.
4. Construct DAG for the given expression.
5. Explain loop optimization techniques with example.
6. Explain any five code optimization techniques in compiler designing with suitable example.
7. What are the different ways of representing Intermediate code? Explain with suitable example.
8. Generate Three address code for given code.
9. Explain Peephole optimization techniques with example.
10. Explain the representation of three address code with example.